Area Between Curves

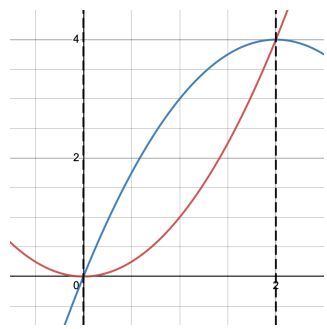
Proposition 1. *The area between the curves $y = f(x)$ and $y = g(x)$ between $x = a$ and $x = b$ is given by:*

$$\int_a^b |f(x) - g(x)|\, dx$$

Author's Note 1. *Problems like these often involve integrating an absolute value symbol, which can be tricky! Remember to sort the interval $[a, b]$ into the regions where $f(x) \geq g(x)$ and $f(x) \leq g(x)$.*

Example 2. *Compute the area between the curves:*

$$f(x) = 4x - x^2 \quad \textit{and} \quad g(x) = x^2$$



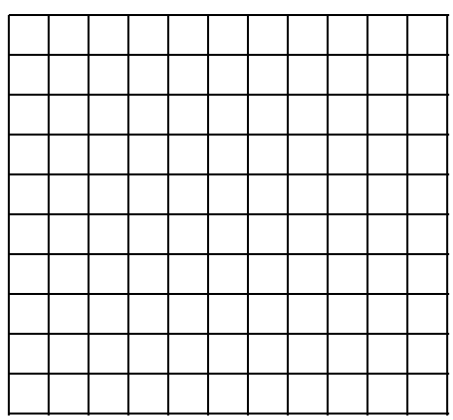
Answer: From the image, we know that the correct integral to set up is:

$$\int_0^2 f(x) - g(x)dx = 2\int_0^2 2x - x^2 dx = 2\left[x^2 - \frac{x^3}{3}\right]_0^2 = 2\left(4 - \frac{8}{3}\right) = \frac{8}{3}$$

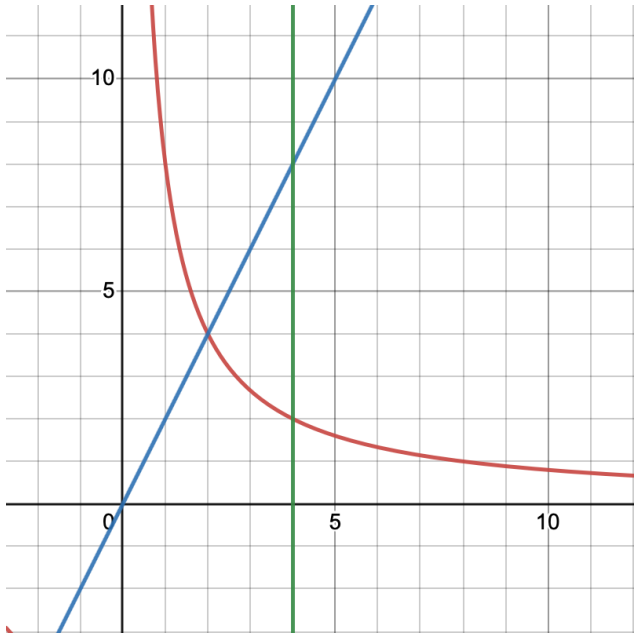
Problem 1. *Compute the area between the curves:*

$$f(x) = \frac{8}{x}, \qquad g(x) = 2x, \qquad \textit{and} \qquad x = 4.$$

Graph the region in the grid below.



Answer: First we graph:

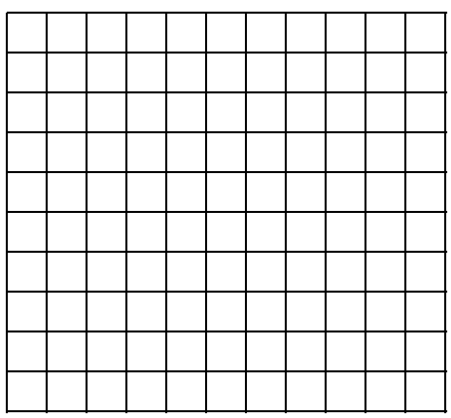


and integrate:

$$\int_2^4 2x - \frac{8}{x} dx = \left[x^2 - 8\ln(x)\right]_2^4$$

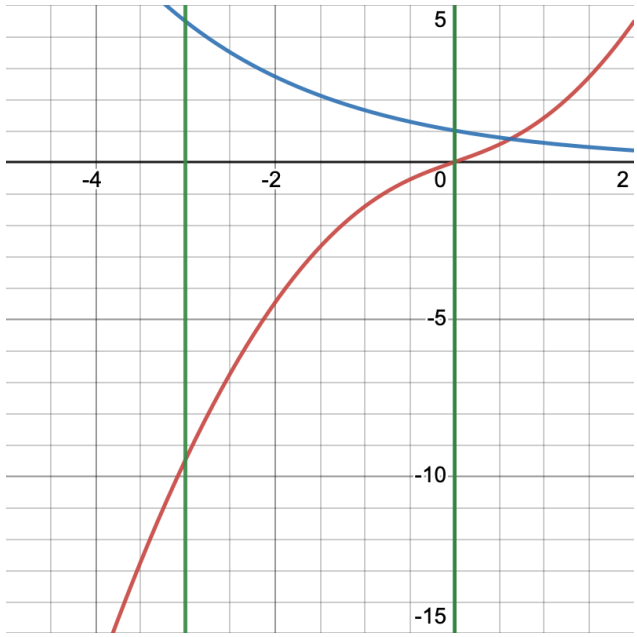
Problem 2. *Sketch the region enclosed by the following curves, and then* ***set up an integral to*** *compute the area of these regions:*

$$y = x\sqrt{x^2+1}, y = e^{-\frac{1}{2}x}, x = -3 \quad \textit{and the} \quad y\textit{-axis.}$$



Answer: Show students here that it's not necessary to know the exact shape of a graph so long as they can

find intersection points and know the region of integration.



The correct integral would be:

$$\int_{-3}^{0} e^{-\frac{1}{2}x} - x\sqrt{x^2+1}dx$$